Assignment No : 06

Title:

Design and development of a dynamic **E-Commerce Web Application for Used Items (ReMarket – Sell & Buy Platform)** using HTML, CSS, JavaScript, and Bootstrap.

Objectives:

1. To understand the complete workflow of developing a real-world e-commerce web application.
2. To implement a responsive and interactive buy/sell platform using frontend technologies like HTML, CSS, JavaScript, and Bootstrap.

Problem Statement:

To design and develop a dynamic **E-Commerce Web Application (ReMarket – Sell & Buy Platform)** where users can:

- Sell their used products (cars, bikes, electronics, etc.)
- Browse and purchase listed items
- View product details with pricing
- Interact with forms for posting and contacting

The system must include **JavaScript-based validation** for forms like product listing and contact form to ensure proper user input.

Outcomes:

Students will be able:

1. To design a professional and responsive e-commerce UI.
2. To implement dynamic features like product listing and form validation.
3. To explore modern frontend frameworks like Bootstrap.

Hardware and Software Requirements:

Hardware:

- Laptop/Desktop
- Minimum 4GB RAM

Software:

- VS Code / Notepad++
- Web Browser (Chrome/Edge)
- HTML5, CSS3, JavaScript
- Bootstrap CDN

Theory :

1) Introduction to JavaScript:

JavaScript is used in this project to add dynamic behavior such as form validation, button interactions, product filtering, and user input handling in the ReMarket platform.

2) JavaScript Concepts Used:

- **Data Types:** String, Number, Boolean (used for user/product data)
- **Functions:** Used for validation and UI interaction

- **Objects:** Represent product details (name, price, category)
- **Events:** Click, submit, hover (for buttons and forms)
- **Strings:** Used to manage input fields and messages

3) Functions, Parameters & Invocation:

- Functions are defined to validate forms like product listing
- Parameters include user inputs (product name, price, contact details)
- Functions are invoked during form submission
- call() and apply() methods help in function execution with context

4) Detailed Website Description (ReMarket Platform)

Homepage (Landing Section):

- Clean and modern UI
- Tagline like *"Buy & Sell Used Items Easily"*
- Call-to-action buttons: **Browse Products / Sell Item**

Navigation Bar:

- Home
- Browse Items
- Sell Product
- Categories
- Contact

Responsive navigation using Bootstrap.

Product Listings Section:

- Displays items like:
    - Cars
    - Bikes
    - Mobiles
    - Electronics
- Each product card includes:
    - Product image
    - Title
    - Price
    - Location
    - “View Details” button

Categories Section:

- Organized browsing (Vehicles, Electronics, Furniture, etc.)
- Helps users quickly filter items

Sell Product Form:

Users can list items by entering:

- Product Name
- Category

- Price
- Description
- Contact Details

JavaScript Validation:

- Checks empty fields
- Ensures price is numeric
- Validates contact details
- Displays error messages if invalid

Product Details Page:

- Shows detailed information about selected product
- Includes price, description, and seller contact

Contact Section:

- User inquiry form
- Email/contact info
- Used for customer support

Footer Section:

- Quick links
- Social media icons
- Copyright

Flowchart of Application:

Start

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Open Website

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Display Homepage

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User Browses Products

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User selects:

→ Buy Product OR Sell Product

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If Sell:

Fill Form → Validate → Submit

If Buy:

View Details → Contact Seller

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End

DESIGN / EXECUTION STEPS:

1. Created HTML structure for homepage, product listings, and forms.
2. Designed UI using CSS and Bootstrap.
3. Implemented responsive layout using Bootstrap grid system.

4. Added JavaScript for form validation and interaction.
5. Integrated all components into a working web app.
6. Tested functionality in browser.

TEST CASES:

Test Case	Input	Expected Output
Valid Product	All correct data	Product listed
Empty Fields	Blank input	Error message
Invalid Price	Text instead of number	Validation error
Navigation Click	Click menu	Page loads correctly
Sell Button	Click	Opens form

Conclusion:

The **ReMarket – Sell & Buy Platform** was successfully developed as a dynamic e-commerce web application. The project demonstrates real-world implementation of product listing, user interaction, and client-side validation using JavaScript. It also highlights responsive design using Bootstrap.